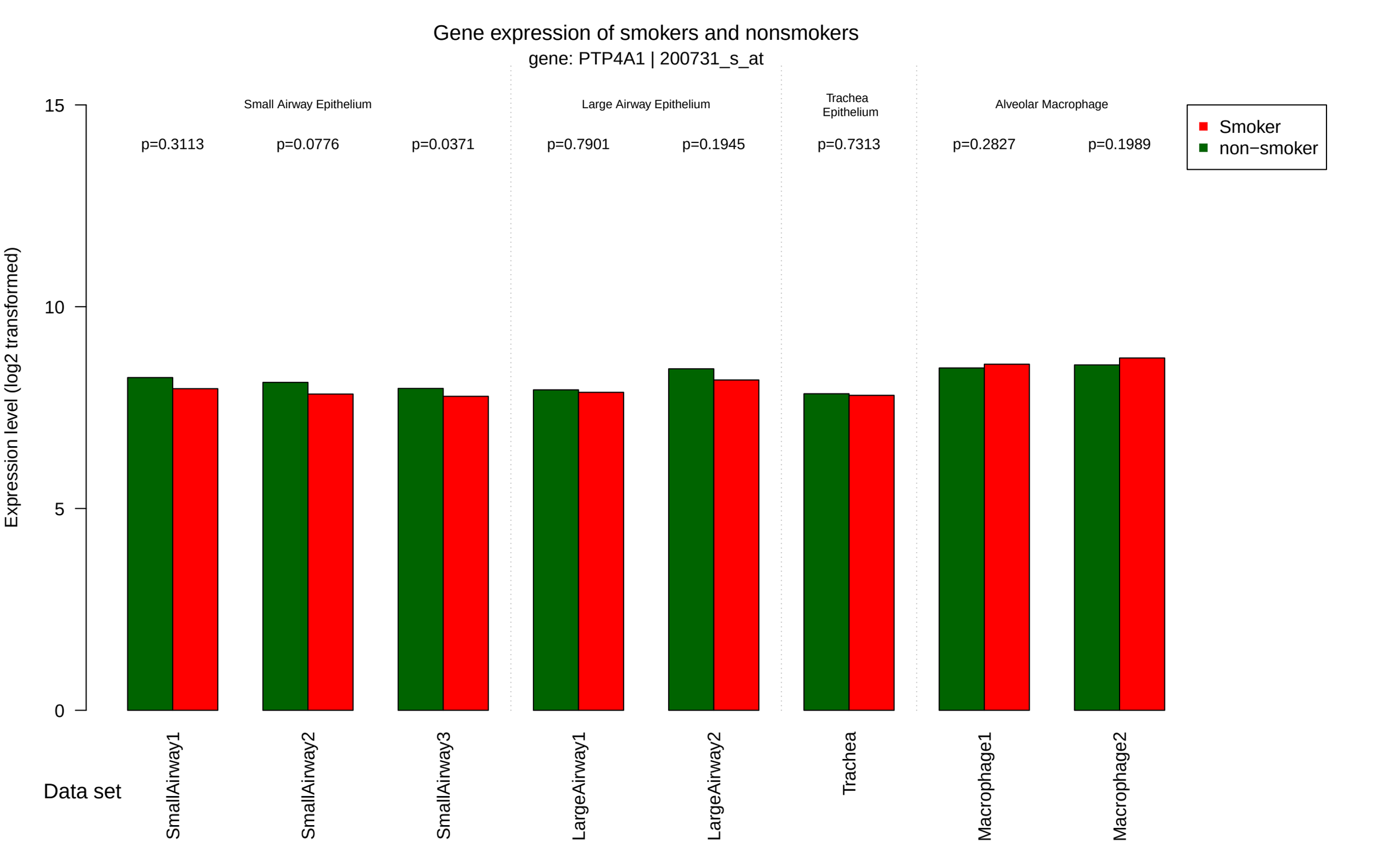
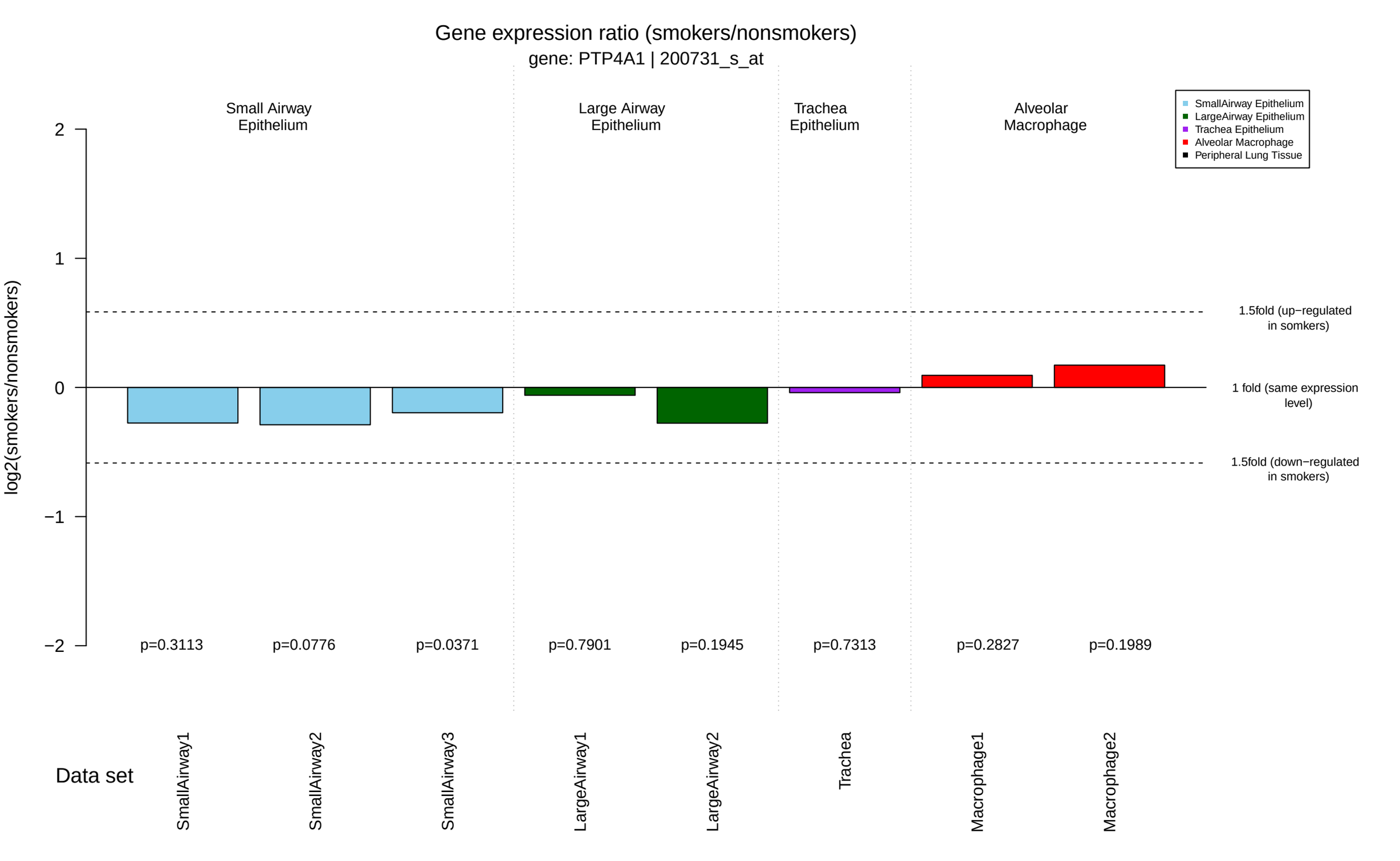
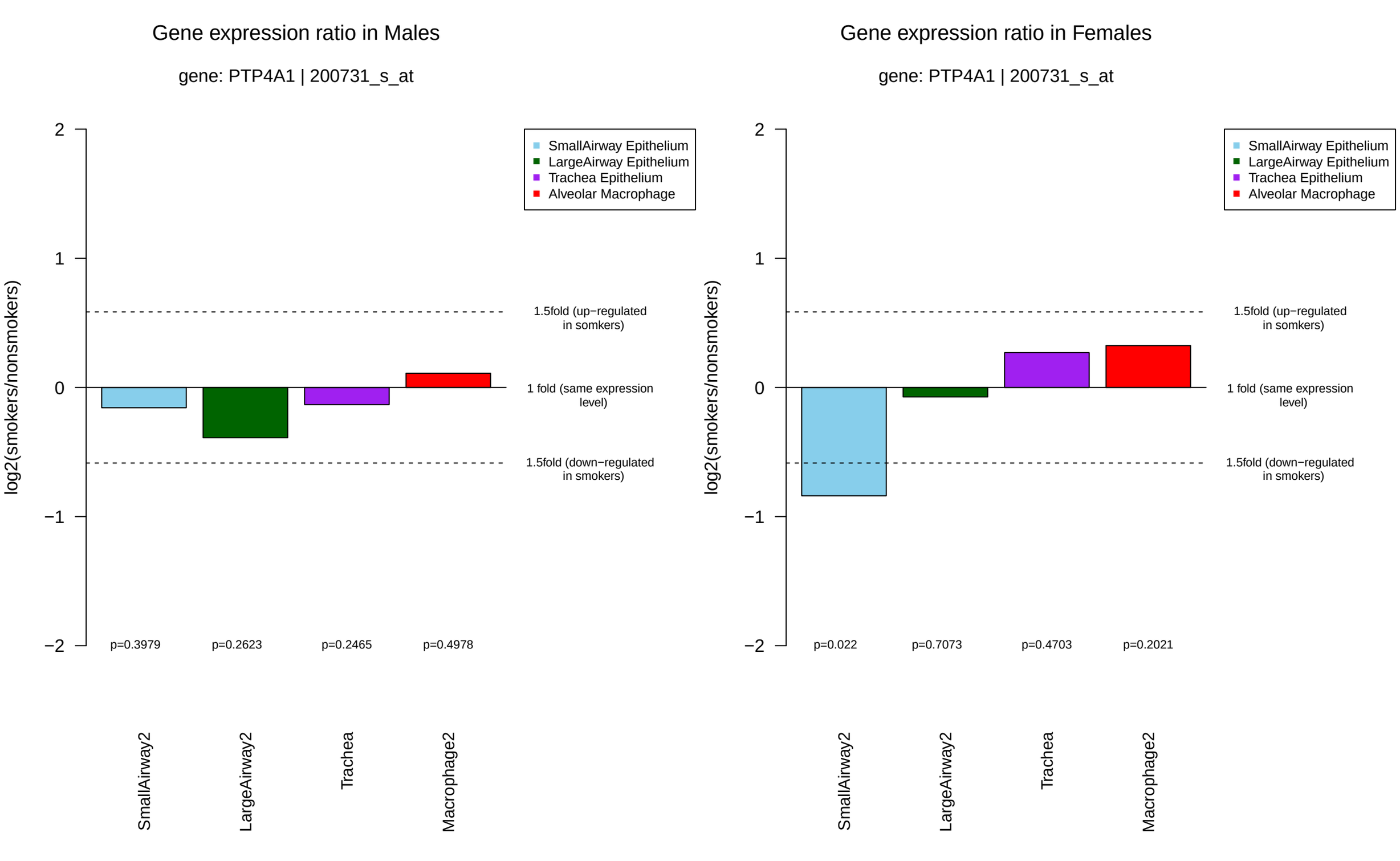
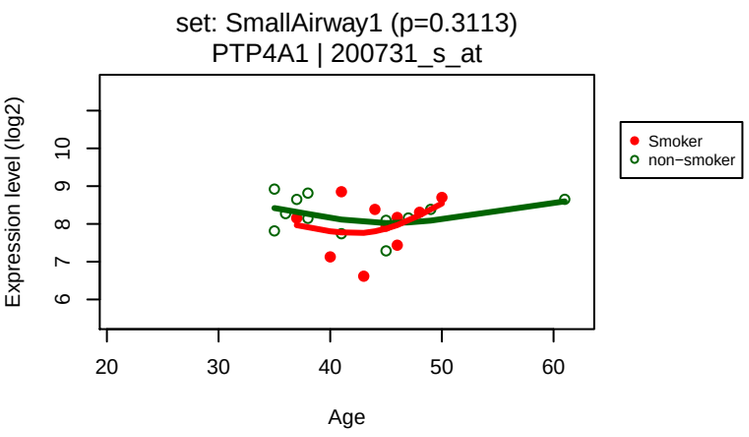
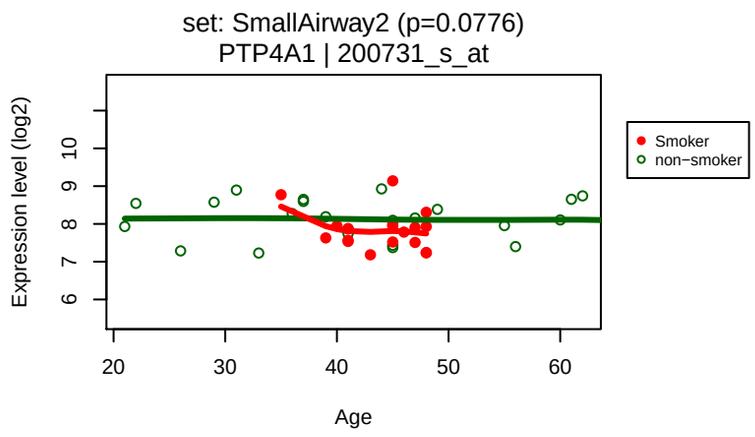
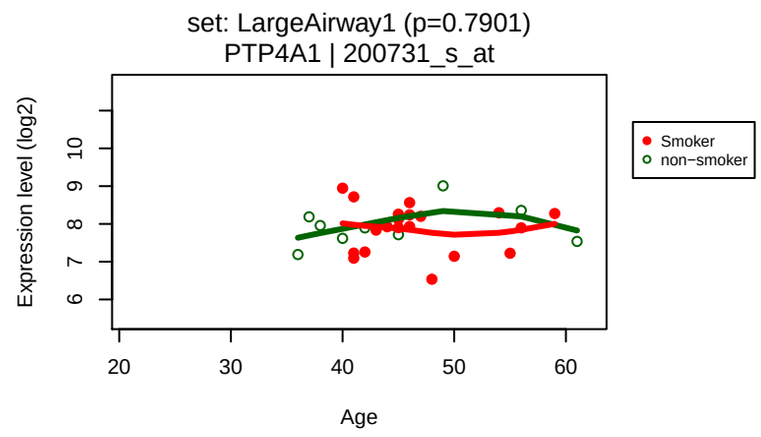

Gene expression of smokers and nonsmokers
gene: PTP4A1 | 200731_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3113
p=0.0776
p=0.0371
p=0.7901
p=0.1945
p=0.7313
p=0.2827
p=0.1989
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

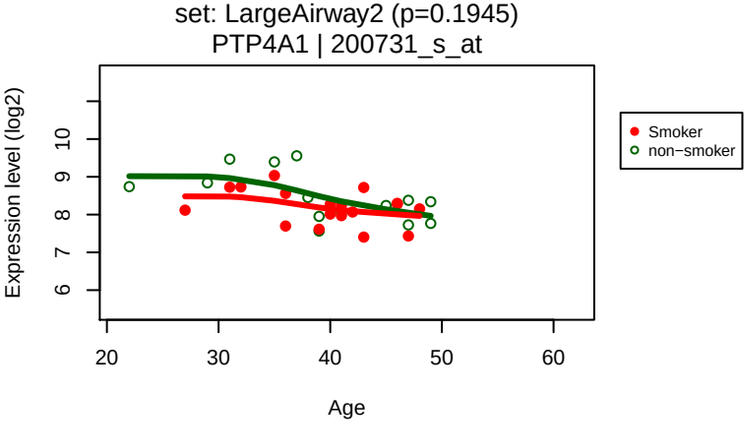

Gene expression ratio (smokers/nonsmokers)
gene: PTP4A1 | 200731_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3113
p=0.0776
p=0.0371
p=0.7901
p=0.1945
p=0.7313
p=0.2827
p=0.1989
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

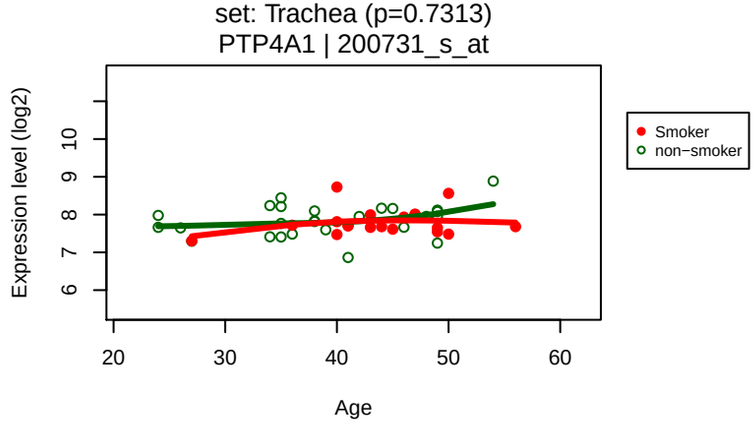

Gene expression ratio in Males
gene: PTP4A1 | 200731_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3979
p=0.2623
p=0.2465
p=0.4978
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTP4A1 | 200731_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.022
p=0.7073
p=0.4703
p=0.2021
SmallAirway2
LargeAirway2
Trachea
Macrophage2

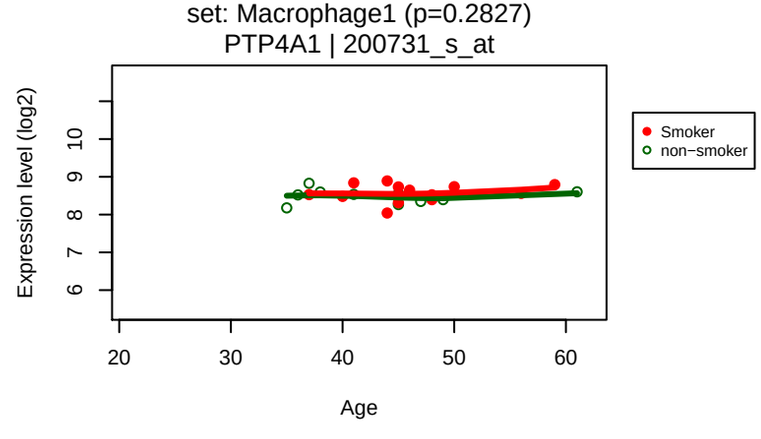

set: SmallAirway1 (p=0.3113)
PTP4A1 | 200731_s_at
Expression level (log2)
Age
Smoker
non−smoker

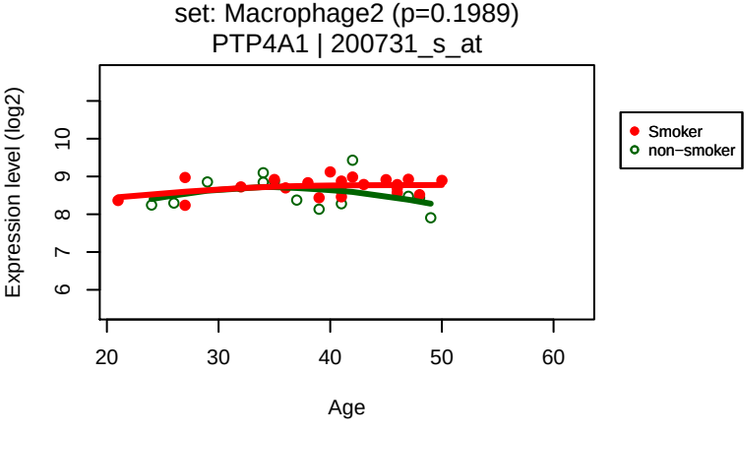

set: SmallAirway2 (p=0.0776)
PTP4A1 | 200731_s_at
Expression level (log2)
Age
Smoker
non−smoker

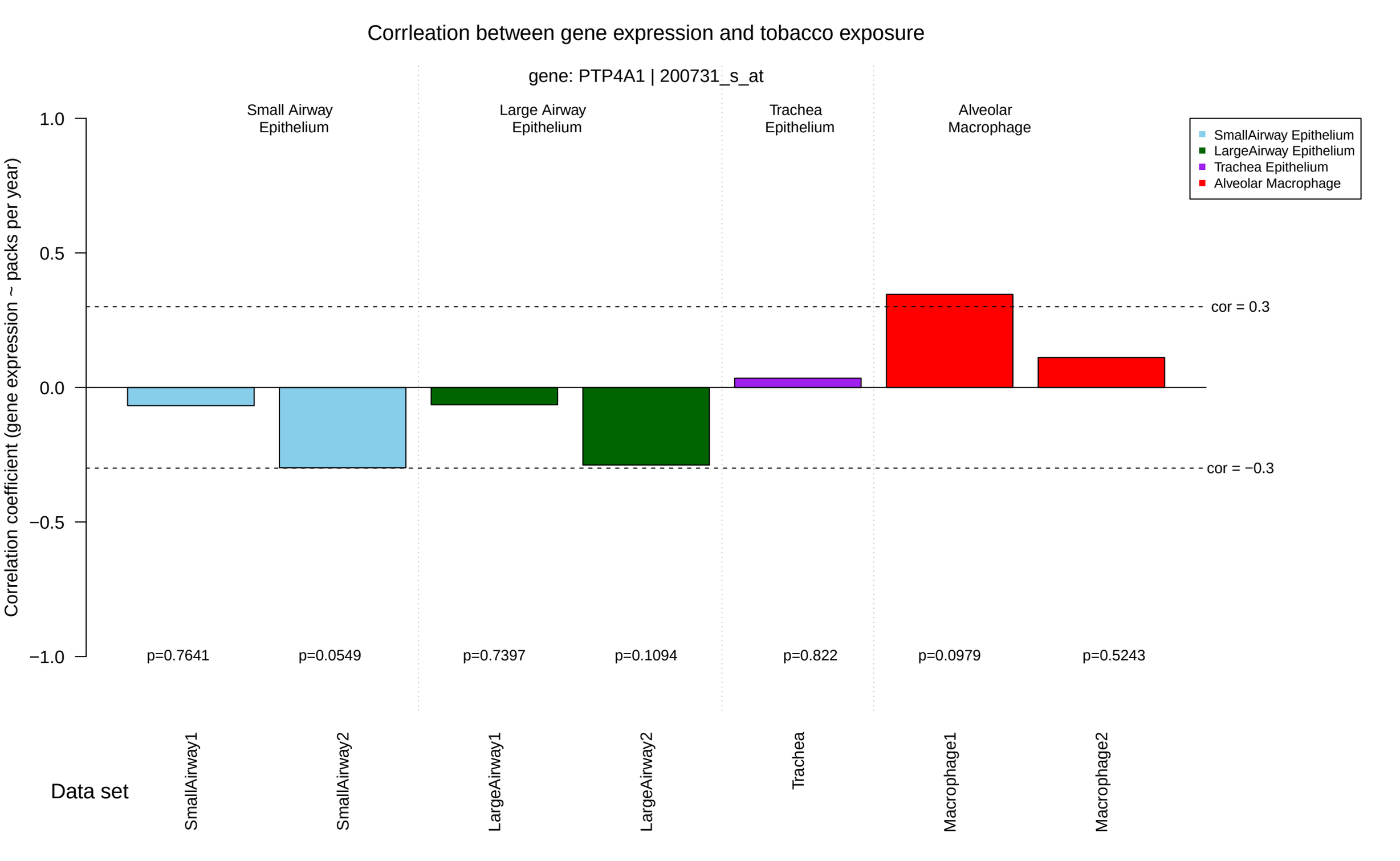

set: LargeAirway1 (p=0.7901)
PTP4A1 | 200731_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1945)
PTP4A1 | 200731_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7313)
PTP4A1 | 200731_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2827)
PTP4A1 | 200731_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1989)
PTP4A1 | 200731_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTP4A1 | 200731_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7641
p=0.0549
p=0.7397
p=0.1094
p=0.822
p=0.0979
p=0.5243
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2